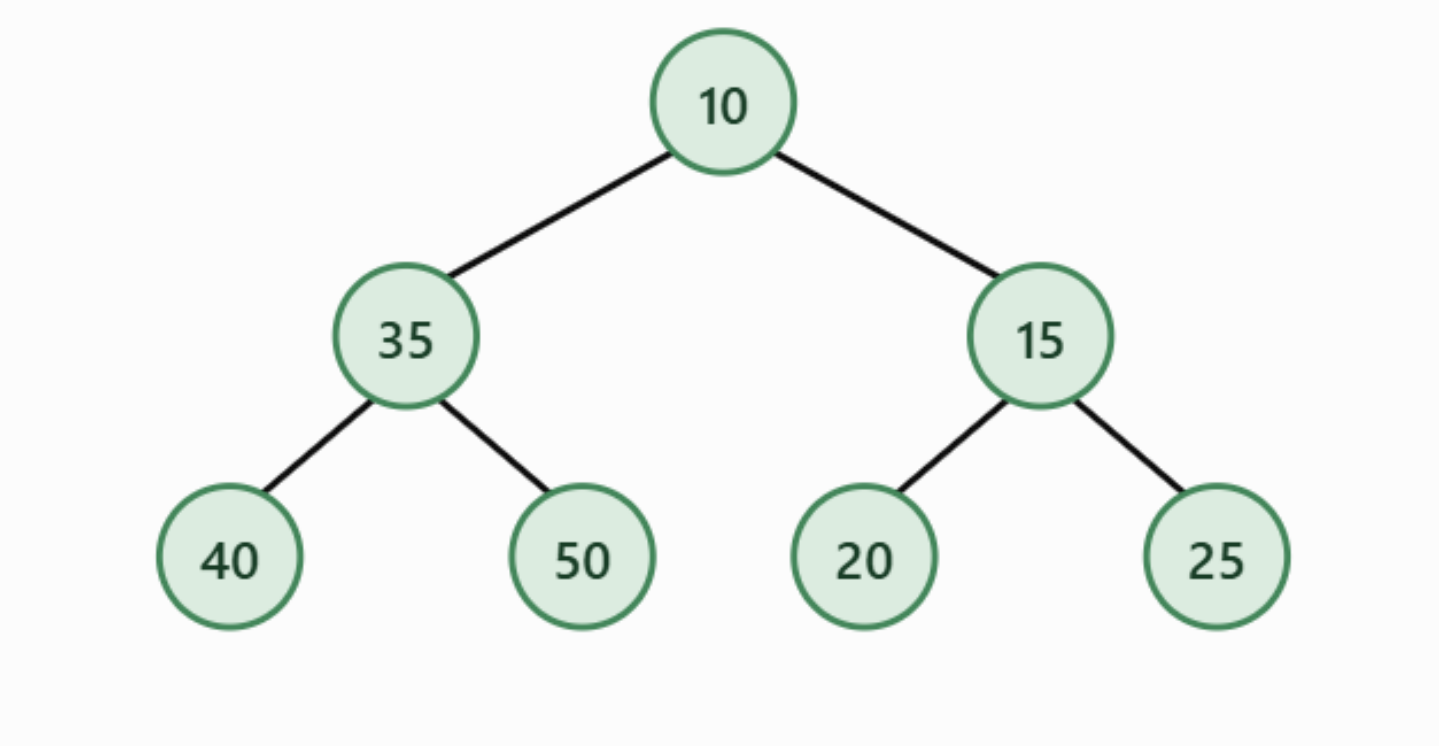
Activity 1: Build a Min Heap and Max Heap

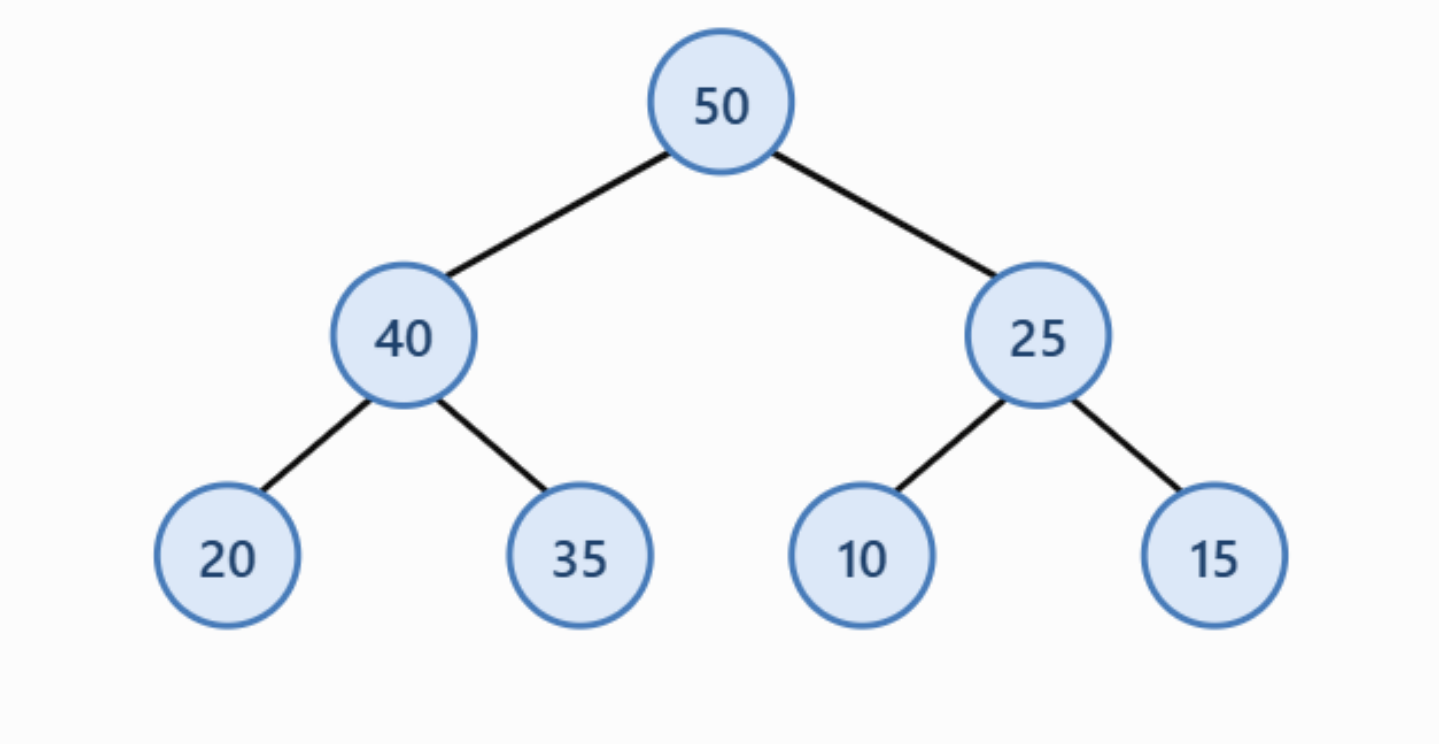
1. **Min Heap**



Array representation: [10, 35, 15, 40, 50, 20, 25]

Root value: 10

2. **Max Heap**



Array representation: [50, 40, 25, 20, 35, 10, 15]

Root value: 50

Objective: Understand the properties of min heaps and max heaps.

3. Difference Between Min Heap and Max Heap

A Min Heap has the smallest number at the top, while a Max Heap has the biggest number at the top. Both arrange numbers in a binary tree, but their rules are different. Basically, Min Heap = smallest first, and Max Heap = biggest first.

Objective: Compare the arrangement of values in both heaps.

Instructions

Given the numbers:

35, 20, 15, 40, 50, 10, 25

1. Insert the numbers one at a time into a min heap.
2. Draw the resulting binary tree.
3. Write its array representation.
4. Repeat the process using a max heap.
5. Identify the root value of each heap.
6. Explain the difference between the two structures.

Expected output: Two heap diagrams, two array representations, and a short explanation.

Activity 2: Heap Operations Challenge

Objective: Practice insertion, extraction, and heapify operations.

Start with this min heap:

[5, 10, 15, 20, 25, 30]

Perform the following operations in order:

1.Insert 3

Insert 3 at the end, then swap it with 15 and 5.

Swaps: 3 ⇔ 15, then 3 ⇔ 5

Heap after operation: [3, 10, 5, 20, 25, 30, 15]

Root value: 3

2. Extract the Minimum

Remove 3, move 15 to the top, then swap it with 5.

Swap: 15 ⇔ 5

Heap after operation: [5, 10, 15, 20, 25, 30]

Root value: 5

3. Insert 8

Insert 8 at the end, then swap it with 15.

Swap: 8 ⇔ 15

Heap after operation: [5, 10, 8, 20, 25, 30, 15]

Root value: 5

4. Extract the Minimum Again

Remove 5, move 15 to the top, then swap it with 8.

Swap: 15 ⇔ 8

Heap after operation: [8, 10, 15, 20, 25, 30]

Root value: 8

Complete the table below.

Operation	Heap array after operation	Root value
Initial heap	[5, 10, 15, 20, 25, 30]	5
Insert 3	**[3, 10, 5, 20, 25, 30, 15]**	**3**
Insert 8	**[5, 10, 8, 20, 25, 30, 15]**	**5**
Extract minimum	**[5, 10, 15, 20, 25, 30]**	**5**
Extract minimum	**[8, 10, 15, 20, 25, 30]**	**8**

Heap operations help keep the smallest value at the top of a Min Heap. When we insert or remove a number, we swap values if needed to keep the heap in the correct order. Basically, we just need to follow the heap rules so the structure stays organized.